Banking & Financial Services

BM

Bank Marketing *Campaign* Analytics

Optimising term-deposit conversion through customer, campaign & macro-economic analytics — across 41,176 direct-marketing contacts.

41,176	11.27%	6.9×
CONTACTS ANALYSED	BASELINE CONVERSION	TOP-DRIVER LIFT

PROJECT
Bank Marketing Campaign Analytics
UCI Portuguese Bank Dataset

SECTOR
Banking & Financial Services
Retail · Direct Marketing

GROUP
Group — 11

SUBMISSION
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SECTION 01

Executive Summary

This project analyses 41,176 direct-marketing contacts from a Portuguese bank's term-deposit campaign to identify the customer, campaign and macro-economic factors that drive subscription. Using descriptive analytics, hypothesis testing and logistic regression, the team produced four interactive Tableau dashboards and five ranked recommendations that together offer a clear path to lift conversion well above the 11.27% baseline.

41,176	11.27%	4,639	6.9×
CONTACTS ANALYSED	BASELINE CONVERSION	SUCCESSFUL CUSTOMERS	TOP-DRIVER LIFT

Three Headline Findings

- **Previous-campaign success** is the single strongest predictor — repeat-success customers convert at 65.11% vs 8.83% for cold contacts (6.9× lift, Chi-Square = 4,114).
- **Macro timing dominates volume.** Low/medium-rate windows yield 24–44% conversion vs 4.83% in high-rate windows. The bank's heaviest contact volume falls in the worst macro window.
- **Call duration signals intent.** Longer calls correlate strongly with subscription (r = +0.41, logistic OR = 3.23). Overcalling (6+ contacts) actively destroys yield, dropping conversion to 5.49%.

Five Ranked Recommendations

- **R1.** Re-rank the call list by prior-success and demographic flags (Senior 55+ / Young ≤25).
- **R2.** Cap campaign call-intensity at 3 contacts per customer — build a soft-stop in the dialler.
- **R3.** Shift channel mix toward cellular (14.74% vs 5.23% landline conversion, ~3× advantage).
- **R4.** Time heavy campaigns to low-rate windows and to March / September / December.
- **R5.** Deploy a lightweight 6-feature propensity score (logistic regression, 90.8% accuracy).

SECTION 02

Sector & Business Context

Retail banking relies on term deposits as a stable funding source. Outbound call-centre campaigns are high-cost per contact and typically produce single-digit conversion rates — at 11.27% baseline, roughly 89 of every 100 calls produce no revenue yet consume full agent time. Even small percentage-point gains compound into significant ROI shifts. The primary stakeholder is the Head of Direct Marketing / Campaign Manager, supported by the analytics function, call-centre operations, Treasury, Compliance, and Branch Managers in high-conversion regions.

SECTION 03

Problem Statement & Objectives

Given historical contact-level records comprising customer demographics, contact metadata, prior-campaign history and macro-economic indicators, identify the factors statistically associated with subscription and quantify their relative impact so the campaign manager can re-prioritise outreach.

Success Criteria

- Reproducible cleaning pipeline yielding a single canonical analysis dataset.
- All major hypotheses tested at alpha = 0.05 with appropriate statistical methods.
- Four interactive Tableau dashboards covering customer, campaign, interaction-history and economic perspectives.
- A logistic-regression baseline with explainable coefficients and accuracy ≥ 85%.
- Concrete, ranked recommendations linked back to specific statistical evidence.

SECTION 04

Data Description

Source: UCI ML Repository — Moro, Cortez & Rita (2014). Captures direct-marketing phone campaigns of a Portuguese banking institution. Raw shape: 41,188 rows × 21 columns. Cleaned shape: 41,176 rows × 27 columns (12 duplicates dropped; 7 features engineered). Target: subscription_flag — 1 = subscribed (11.27%); highly imbalanced. Key columns: age, job_type, contact_type, call_duration_sec, num_contacts_campaign, previous_campaign_outcome, interest_rate, employment_variation_rate, CCI, subscription_flag. Macro indicators (EVR, CPI, CCI, Euribor 3M, nr_employed) span the 2008 financial-crisis aftermath.

Data Quality: No null values. 12 duplicate rows removed. Outliers in num_contacts_campaign capped at 10. Value 999 in days_since_last_contact (meaning "never contacted") replaced with categorical contact_recency. "Unknown" values in demographic fields retained as a valid segment — their conversion behaviour differs measurably from fully-profiled customers. No native region field; NUTS II added via probability-based imputation conditioned on job_type.

SECTION 05

Cleaning & Transformation

The 17-step pipeline produces Bank_Marketing_Dataset_Cleaned_Data.csv (41,176 rows × 27 columns) used by all downstream analysis and dashboards.

Step	Action	Detail
1–2	Load & De-duplicate	Loaded semicolon-separated file; dropped 12 fully duplicate rows → 41,176 unique records.
3–5	Type & Value fixes	Cast nr_employed to int; renamed cryptic columns; standardised yes/no → 1/0; Unknown casing fixed.
6–7	Category cleanup	Merged tiny "illiterate" into basic.4y; applied explicit category dtypes for Tableau.
8	Cap outliers	num_contacts_campaign clipped at 10.
9–13	Feature engineering	age_group (5 buckets), call_intensity, contacted_before, contact_recency (replaced 999-day placeholder), prev_success_flag.
14–16	More engineering	rate_environment (Low/Med/High), education_ordinal (hierarchy mapping), region NUTS II (prob. imputation from job_type).
17	Output	41,176 × 27 CSV — all fields Tableau-friendly and self-explanatory.

KPI Framework

Each dashboard surfaces headline KPIs. Values computed on the full cleaned dataset (n = 41,176).

KPI	Value	Formula / Rationale
Conversion Rate	**11.27%**	subscribed / contacts × 100 — headline benchmark for all segments
Successful Customers	**4,639**	Count(subscription_flag = 1) — deposit-book growth
Avg Call Duration	**4.3 min**	Mean(call_duration_sec) — strongest correlate (r = +0.41)
Avg Contacts / Customer	**2.44**	Mean(num_contacts_campaign) — crossing 3 calls reduces yield
Cellular Conversion	**14.74%**	subscribed / cellular contacts — ~3× landline rate
Prior-Success Conversion	**65.11%**	subscribed / contacts where prev = success — 6.9× lift
Cold-Contact Conversion	**8.83%**	subscribed / never-contacted — floor benchmark
Low-Rate Env. Conversion	**24.02%**	subscribed / contacts in low-rate window — ~5× high-rate
High-Rate Env. Conversion	**4.83%**	Worst-case macro benchmark
Avg Interest Rate	**3.62%**	Mean(interest_rate) — macro context indicator
Avg Consumer Conf. Index	**-40.50**	Mean(CCI) — sentiment proxy; positively correlates (r = +0.05)

KPI-to-Dashboard Mapping

Dashboard	Headline KPIs
Customer Profile	Total Customers, Customers Contacted, Successful, Unsuccessful, Success Rate
Campaign Effectiveness	Conversion Rate, Avg Call Duration, Avg Contacts Per Customer, Cellular Conversion
Interaction History	Previously Contacted, Prior Success Count, Prior-Success Conversion, Cold-Contact Conversion
Economic Impact	Low-Rate Conv., High-Rate Conv., Avg Interest Rate, Avg Consumer Confidence Index

SECTION 07

Exploratory Analysis

11.27%	U-Shaped	14.74%	50.55%
BASELINE CONV.	AGE PATTERN	CELLULAR CONV.	MARCH CONV.

Major Trends

- **Class imbalance dominates.** Only 11.27% subscribe — every analysis accounts for the rare-event nature of the target.
- **Age is U-shaped.** Senior 55+ converts at 46.9% and Young ≤25 at 21.0%, while the 36–45 middle cohort lags at 8.4% — the largest segment by volume is the worst by yield.
- **Volume vs yield mismatch.** May has the highest contact volume but March (50.55%), September (44.91%) and December (48.90%) produce the highest conversion rates.
- **Cellular >> telephone.** 14.74% vs 5.23% — nearly a 3× channel advantage.
- **Students and retired customers** convert at materially higher rates than blue-collar workers.
- **Overcalling hurts.** Conversion drops monotonically: 13.04% (1 call) → 11.46% (2 calls) → 9.83% (3–5) → 5.49% (6+ calls).
- **Prior-success customers** are 6.9× more likely to subscribe again (65.11% vs 9.4%).
- **Macro environment is decisive.** Low/medium-rate windows yield 5–9× the conversion of high-rate windows.
- **Outlier notes:** num_contacts_campaign capped at 10; call_duration_sec outliers retained (genuine, highly predictive); macro indicators highly inter-correlated.

SECTION 08

Statistical Analysis

Six hypotheses tested at alpha = 0.05 using ANOVA (3+ group comparisons) and Chi-Square Tests of Independence (categorical pairs). Effect sizes quantified with Pearson correlation and a multivariate logistic regression (6 features, 90.8% accuracy).

Factor	Test	Statistic	p-value	Result
Age Group	ANOVA	281.01	< 0.001	**SIGNIFICANT**
Job Type	Chi-Square	961.74	< 0.001	**SIGNIFICANT**
Call Intensity	ANOVA	63.01	< 0.001	**SIGNIFICANT**
Contact Type	Chi-Square	862.08	< 0.001	**SIGNIFICANT**
Previous Success	**Chi-Square**	**4,113.95**	**< 0.001**	**HIGHLY SIG.**
Rate Environment	ANOVA	1,977.46	< 0.001	**SIGNIFICANT**

Pearson Correlation with Subscription Outcome

Feature	r	Interpretation
call_duration_sec	+0.405	Strongest positive — behavioural intent signal
consumer_conf_idx	+0.055	Modest positive — sentiment proxy
age	+0.030	Small positive — driven by Senior 55+ cohort
num_contacts_campaign	-0.071	Negative — overcalling hurts
employment_var_rate	-0.298	Strong negative — contracting economy boosts deposits
interest_rate	-0.308	Strongest negative — high rates suppress uptake

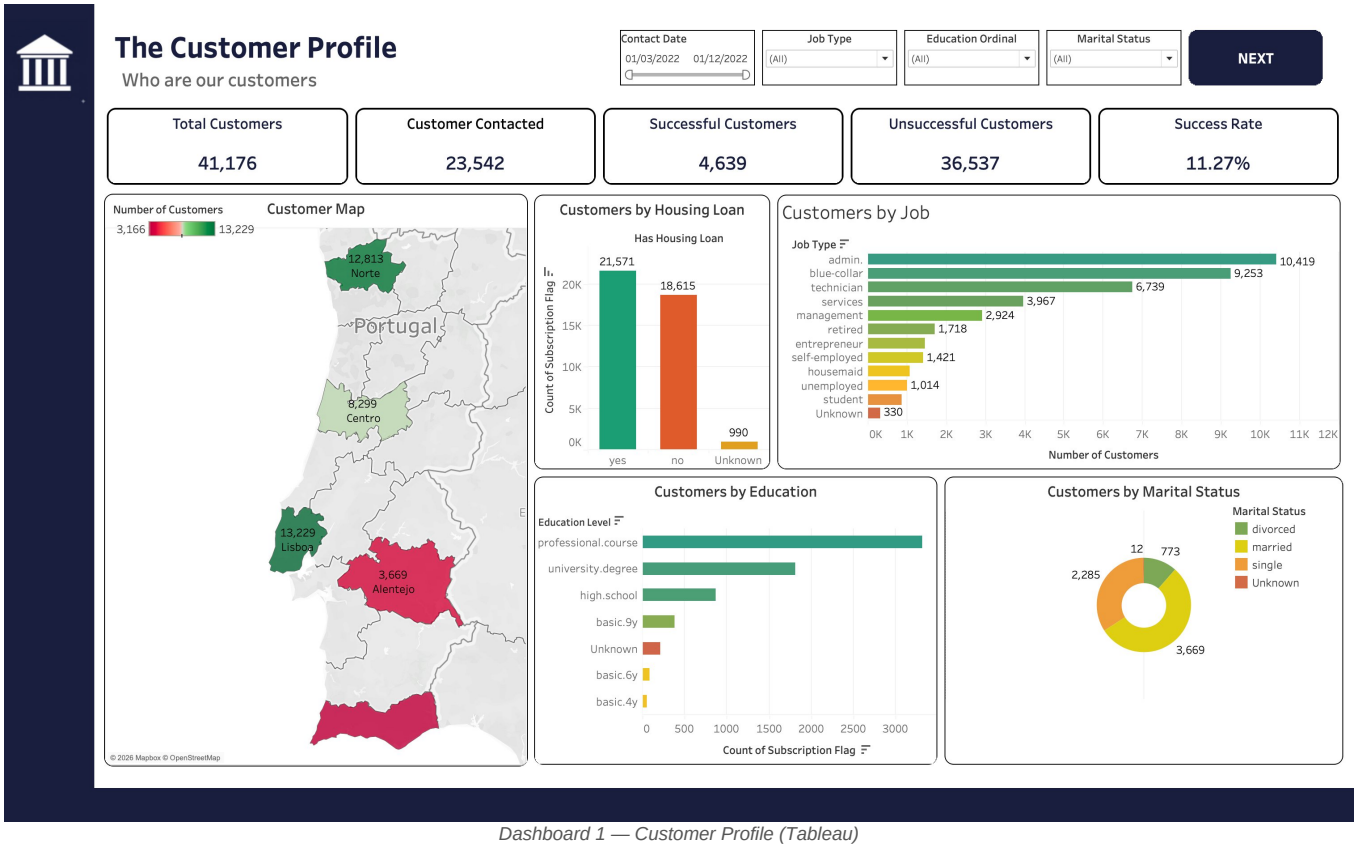
Logistic Regression Key Odds Ratios (90.8% accuracy, 6 features)

- **Call duration (OR = 3.23)** — triples the odds; train agents to sustain qualified conversations.
  - **Interest-rate environment (OR = 0.35)** — high rates cut odds ~65%; ramp campaigns in low-rate windows.
  - **Prior-success flag (OR = 1.39)** — 6.9x lift; should be first-sort on every dialling list.
  - **num_contacts_campaign (OR = 0.91)** — negative coefficient confirms calling more hurts, not helps.

Dashboard 1 — Customer Profile

Who are our customers? Demographic distribution by region, job, education and marital status.

41,176	23,542	4,639	36,537	11.27%
TOTAL CUSTOMERS	CONTACTED	SUCCESSFUL	UNSUCCESSFUL	SUCCESS RATE



Dashboard 1 — Customer Profile (Tableau)

Key Observations

- Norte (12,813) and Lisboa (13,229) together account for >63% of all customers — primary target regions.
- Admin (10,419) and blue-collar (9,253) are largest segments; retired and student customers carry far higher conversion rates, creating a clear targeting opportunity.
- Housing loan holders (21,571) outnumber non-holders (18,615) — cross-sell potential worth investigating.
- Professional.course and university.degree dominate — messaging should reflect product sophistication.
- 11.27% success rate confirms the default strategy is undifferentiated; propensity scoring would concentrate effort on highest-yield segments.
- Married customers form the largest marital segment; household financial stability may explain greater willingness to lock funds into deposits.

Dashboard 2 — Campaign Effectiveness

How well are we calling? Subscription rate by call intensity, contact channel, month and prior outcome.

11.27%	4.3	2.44	14.74%
CONVERSION RATE	AVG CALL DURATION	AVG CONTACTS / CUST.	CELLULAR CONV.

Dashboard 2 — Campaign Effectiveness (Tableau)

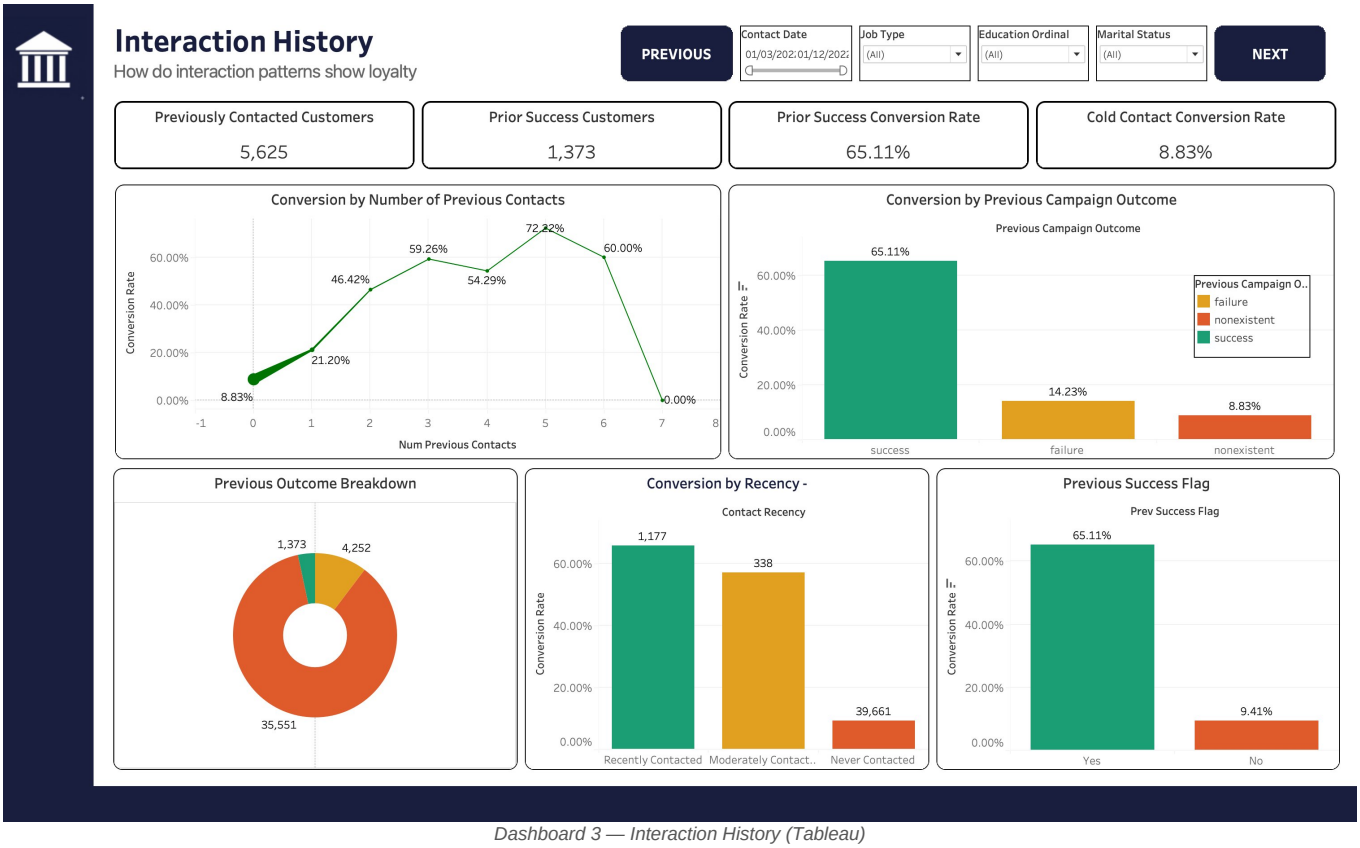
Key Observations

- **Call intensity inversely correlated with conversion:** 13.04% at 1 call → 5.49% at 6+ calls. Every call beyond the third destroys yield.
- **March (50.55%), December (48.90%), September (44.91%)** are top months — yet May is the highest-volume month at only 6.44%. Classic volume/yield mismatch.
- **Cellular at 14.74% vs telephone at 5.23%** — a ~3× advantage from simply changing dialler configuration.
- Prior-outcome breakdown: 35,551 never contacted, 4,252 prior failures, only 1,373 prior successes — yet that 1,373 drives a disproportionate share of all subscriptions.
- Recently contacted customers (1,177 successes) convert at substantially higher rates than never-contacted groups — recency of relationship matters.
- Avg call duration of 4.3 min — longer calls strongly correlate with subscription (r = +0.41); training agents to sustain qualified conversations could directly lift conversion.

Dashboard 3 — Interaction History

How do interaction patterns show loyalty? Prior campaign history, recency and success flags as predictors.

5,625	1,373	65.11%	8.83%
PREVIOUSLY CONTACTED	PRIOR SUCCESS	PRIOR-SUCCESS CONV.	COLD-CONTACT CONV.



Dashboard 3 — Interaction History (Tableau)

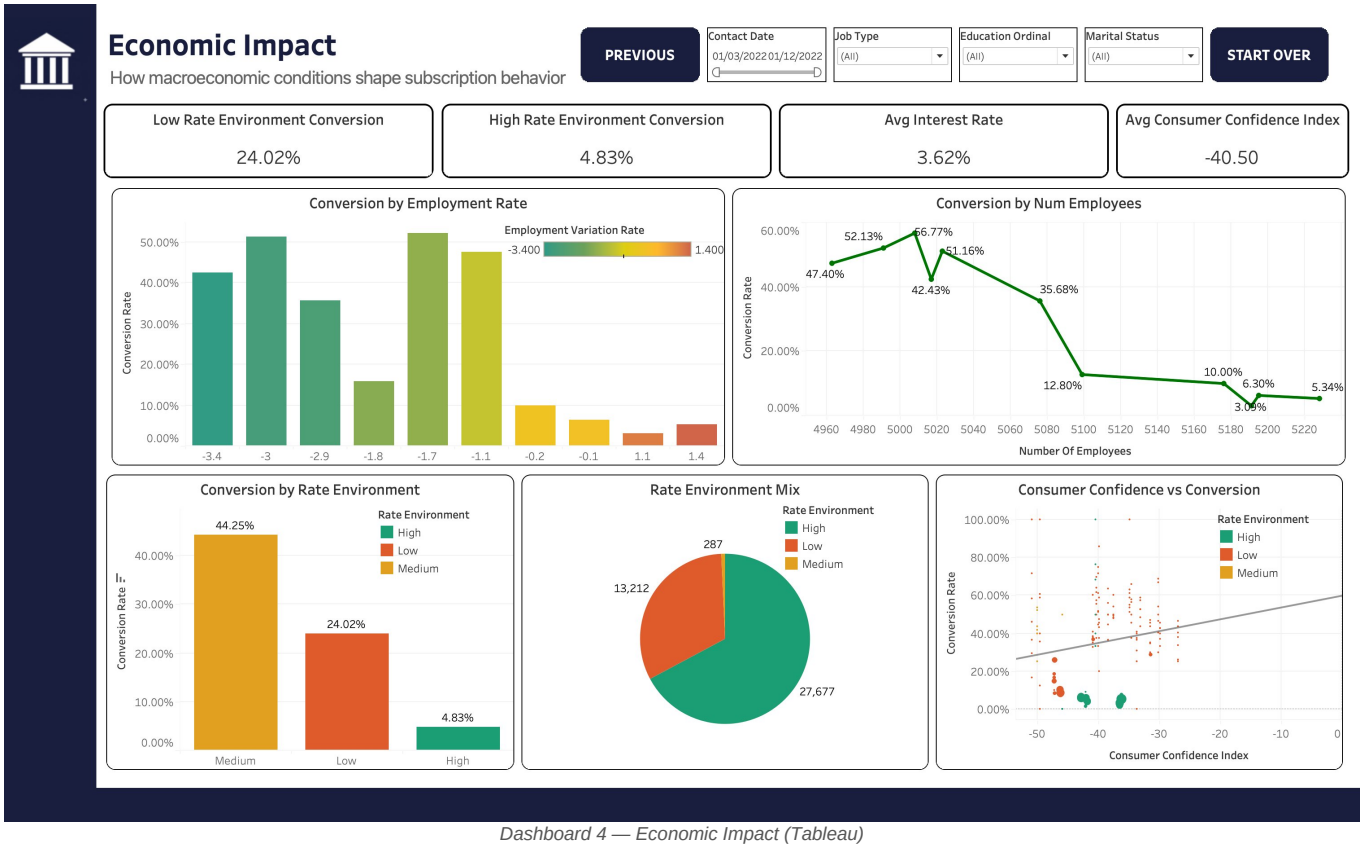
Key Observations

- **Prior success is the strongest predictor** — 65.11% vs 8.83% for cold contacts (6.9× lift). Highest-leverage segment in the dataset.
- **Prior failure still beats cold contact:** 14.23% vs 8.83% — even a failed prior relationship has re-activation value worth pursuing.
- **Conversion peaks at 5 previous contacts (72.22%)** then drops sharply — relationship depth matters up to a natural ceiling.
- **Contact recency drives conversion** — recently contacted customers outperform never-contacted groups substantially.
- **Previous Success Flag binary divide:** flag = 1 (65.11%) vs flag = 0 (9.41%) — ~7× gap. This single binary field should be the first-sort on every campaign dialling list.
- Only 1,373 of 41,176 (3.3%) carry a prior-success flag — this elite segment is significantly under-prioritised relative to its conversion power.

Dashboard 4 — Economic Impact

How do macroeconomic conditions shape subscription? Rate environment, employment trends and consumer sentiment.

24.02%	4.83%	3.62%	-40.50
LOW-RATE ENV. CONV.	HIGH-RATE ENV. CONV.	AVG INTEREST RATE	AVG CONSUMER CONF.



Dashboard 4 — Economic Impact (Tableau)

Key Observations

- **Rate environment is the most powerful macro lever:** Medium 44.25% → Low 24.02% → High 4.83%. A ~9× swing between best and worst macro context.
- **High-rate periods dominate contact volume** (27,677 vs 13,212 in low-rate) — the bank calls most when customers are least receptive. Major structural inefficiency.
- **Employment variation rate strongly predicts conversion** — at EVR = –3.4, conversion reaches 42–52%. Customers in uncertain economies flock to safe deposits.
- **Higher employee counts (5,140+) correlate with lower conversion** (5.34% at 5,220) — large workforce periods coincide with economic expansion when deposits are less attractive.
- **Consumer confidence positively correlates with conversion** — less negative CCI (closer to 0) means higher willingness to commit. Medium rate environment (287 contacts, 44.25%) is a tiny but extraordinarily productive window.
- Macro indicators (EVR, CPI, interest rate, employee count) are mutually correlated — any one can serve as a reliable proxy for campaign timing decisions.

Key Insights

Twelve decision-oriented findings, each tied to a specific statistical result or KPI.

01 Baseline conversion is 11.27% — every recommendation is benchmarked against this rate.

02 Previous-campaign success is the single strongest predictor. Repeat-success customers convert at 65.11% vs 8.83% cold (Chi-Square = 4,113.95, 6.9× lift).

03 Conversion is U-shaped in age. Senior 55+ converts at 46.9%; Young ≤25 at 21.0%; the 36–45 majority lags at 8.4% — largest segment, worst yield.

04 Students and retired customers dominate by occupation yield; blue-collar workers underperform the overall average.

05 Cellular outperforms landline ~3× (14.74% vs 5.23%, Chi-Square = 862.08). Channel choice is one of the most actionable levers.

06 Overcalling destroys yield. Conversion drops from 13.04% (1 call) to 5.49% (6+ calls). Logistic OR on num_contacts_campaign is negative.

07 March, September and December are top months (50.55%, 44.91%, 48.90%) — but May is highest-volume at only 6.44%. Clear volume/yield mismatch.

08 Macro-economic regime is decisive. Low-rate windows: 24.02% vs 4.83% high-rate (F = 1,977.46). Bank calls most in worst macro window.

09 Call duration is the strongest behavioural signal (r = +0.41; logistic OR = 3.23) — qualified longer calls drive conversion.

10 Contact recency matters — recently contacted customers outperform cold leads regardless of raw days since last contact.

11 Macro indicators (EVR, CPI, interest rate, employee count) are highly inter-correlated — any one serves as a reliable timing proxy.

12 Logistic-regression baseline reaches 90.8% accuracy with just 6 features — a lightweight propensity model is immediately feasible with no new data.

Business Recommendations

Five ranked recommendations, each anchored to a specific statistical result. Priority reflects expected business impact relative to implementation cost.

R1 Re-rank the call list by prior-success & demographic flags

Move prior-success customers and Senior 55+ / Young ≤25 cohorts to the top of every daily call list.

Evidence: Evidence: 6.9× lift on prior success (Chi-Square = 4,114); 4.2× lift on Senior 55+ vs Mid 36–45.

Expected impact: Lift expected conversion in the re-contact tier from ~11% toward 30–50% on targeted segments; reduces wasted calls on the 36–45 majority.

R2 Cap campaign call-intensity at 3 contacts per customer

Build a soft-stop into the dialler — each marginal call after the third reduces yield.

Evidence: Evidence: 13.04% → 5.49% by call 6+. Logistic OR on num_contacts_campaign is negative.

Expected impact: Reduces agent hours on low-yield repeat calls; improves customer experience and compliance posture.

R3 Shift channel mix toward cellular

Phase down landline outreach where cellular contact is available.

Evidence: Evidence: Cellular converts ~3× landline (14.74% vs 5.23%, Chi-Square = 862).

Expected impact: Direct uplift on conversion rate; potential cost savings if cellular dialling is cheaper per minute.

R4 Time campaigns to low-rate windows and to March / September / December

Run heavy campaigns when interest_rate is in the Low band and during the three highest-yield months.

Evidence: Evidence: Macro timing is associated with a ~9× swing in conversion rate (F = 1,977.46).

Expected impact: Even partial alignment to low-rate windows is highly leveraged given the enormous swing between best and worst macro context.

R5 Deploy a lightweight propensity score using the 6 logistic-regression features

Productionise the model (exclude post-hoc call_duration; use the other 5 as a pre-call score) and feed it into the dialler.

Evidence: Evidence: 90.8% accuracy on the cleaned dataset with no new data dependencies.

Expected impact: Gives the campaign manager a reproducible, explainable scoring layer immediately.

Limitations & Next Steps

Data Limitations

- Public benchmark dataset — findings reflect one Portuguese bank's historical campaigns and may not generalise to other geographies, products or eras.
- No native region field — NUTS II column is probability-imputed from job_type; regional dashboards are illustrative, not ground-truth geography.
- "Unknown" as a demographic category in job_type, marital_status, education_level, credit_default and loan fields limits precision of segmentation analysis.
- Severe class imbalance (11.27% positives) — accuracy alone is misleading; precision/recall/F1 needed for a deployable model.
- Capping num_contacts_campaign at 10 trades a small amount of tail signal for stable aggregations.

Method Limitations

- Logistic regression assumes additive log-odds with no feature interactions — tree-based models (XGBoost, LightGBM) would likely outperform.
- call_duration_sec is post-hoc — it cannot be used in a pre-call propensity score despite being the strongest correlate.
- Macro features are highly inter-correlated, inflating standard errors on individual coefficients.
- Hypothesis tests confirm association, not causation — uplift must be established with controlled A/B testing before full rollout.

Suggested Future Work

- Replace the logistic baseline with a gradient-boosted model (XGBoost / LightGBM) and report precision-recall metrics with class-weighted loss.
- Add channel cost per contact to compute a true cost-per-acquisition KPI rather than raw conversion rate.
- Run an A/B test on the call-cap-at-3 recommendation before full rollout — lowest-risk, highest-confidence change.
- Incorporate customer-level transaction data to add wealth and engagement signals beyond demographics.
- Build a genuine region field from address data; deprecate the imputed NUTS II column.
- Deploy the propensity score as a daily-refreshed scoring layer back into the dialler.

Contribution Matrix

Owner = primary responsibility; Support = active contributor; — = no involvement. Details are verifiable through GitHub Insights, PR history, and submitted artifacts.

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT
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Kasula Lalithendra	Support	**Owner**	**Owner**	Support	Support	Support	Support
Jagruthi Pulumati	**Owner**	**Owner**	Support	Support	**Owner**	Support	**Owner**
Sachin Jaiswal	Support	**Owner**	Support	Support	Support	Support	**Owner**
Ankit Raj Singh	Support	Support	Support	**Owner**	Support	Support	Support
Varun Sharma	—	Support	Support	Support	—	Support	—

Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artifacts.

Team Lead: Jagruthi Pulumati **Date:** 29 April 2026